

Insert a Duvet into Its Cover

SOP (1x Episode: one duvet, one cover, at the bed)

One episode puts one duvet all the way inside one cover. The two far corners are matched first. Then the fabric is worked through to the far end. Then the whole thing is shaken so the duvet drops into the corners. Then the buttons are closed. Then it is spread flat on the bed. The episode ends when the duvet sits square inside the closed cover and the cover lies flat and even on the bed. Never split the filling and the spreading into two episodes.

The base is passive. It has no motor. You push it by hand to the side of the bed and lock it there. Nothing is carried away to a table. Everything the episode touches is already there when recording starts: the bed, the bare mattress with its sheet on, the duvet folded at the foot of the bed, and the cover folded next to it.

The bed is used as found. The mattress has its fitted sheet on and nothing else on top. The duvet lies folded at the foot of the bed. The cover lies folded next to it, right side out, with its buttons open and its opening toward the foot of the bed. The episode ends with the duvet inside the cover, the buttons closed, and the cover spread flat over the whole mattress.

This is an in situ task. Four rules come from that. First, all the work happens on the bed itself. You never lift the duvet or the cover up in the air and hold it there. You keep the fabric on the mattress and slide it along the mattress. Second, you work from the head end to the foot end and never the other way. The two far corners are the head corners. They are matched first and they are never let go until the fabric is through. Third, the bed never moves. You never lean on, push, drag, or lift the bed frame or the mattress. Fourth, nothing goes on the floor. Fabric that touches the floor is dirty and ends the episode.

The fabric goes on the bed and nowhere else. The duvet and the cover are big and light, so they slide easily. You hold a corner all the way closed before you pull. A corner that slips out is a lost grip and you take it again. Fabric on the floor is a fault. You do not pick it up and go on. You report it.

The order never changes: lay both out flat; match the two far corners; work the fabric through to the near end; shake it to seat the duvet; close the buttons; spread it on the bed.

The grippers share the work. One holds a corner still while the other pulls or feeds. Only one hand pulls at a time. Nothing is passed from one gripper to the other in the air.

Setup

Do the base positioning and both checklists before you start an episode.

Cell configuration

cell_type: bimanual

Base positioning

The base is passive. It has no motor. You push it into place by hand. You never steer it, nudge it, or move it after recording starts. You park it once, before recording, and it stays there until the episode is over.

1. Push the base by hand up to the long side of the bed. Stop it square to the side edge of the mattress, so the edge runs straight across the frame and both ends look the same distance away.
2. Stop it at the middle of the long side, so the head end and the foot end of the mattress are the same distance from the middle of the base.
3. Stop it close enough that a gripper reaches the middle of the mattress without stretching the arm. Stop it far enough that no arm, wrist, or part of the base touches the bed frame or the mattress edge while both arms work.
4. Check the reach at the head end. With the base parked, a gripper reaches both far corners of the mattress and closes on them without stretching.
5. Check the reach at the foot end. With the base parked, a gripper reaches the folded duvet and the folded cover at the foot of the bed without stretching.
6. Check the sweep. Both arms move from the foot end to the head end over the mattress without hitting each other and without hitting the headboard.
7. Lock the base. Push it once, hard, by hand. It must not roll, slide, or turn.
8. If any line from 1 to 6 fails, push the base to a new spot by hand and start again at line 1. Do not work at a bed the arms cannot reach easily.

The base stays locked and still for the whole episode. Nothing moves it. No arm leans on the bed hard enough to push it. Nobody touches it by hand. You never move it during the task. A base that moves after recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera points at the middle of the mattress. The frame shows the whole mattress, the head end, the foot end, the folded duvet, and the folded cover.
3. The camera sees the mattress from the side, so you can see where the fabric lies and how far it reaches.
4. The camera sees both far corners of the mattress, so you can see a corner matched and see it slip if it slips.
5. The camera sees the opening of the cover and its buttons, so you can see the buttons open and see them closed.

6. The camera sees the top face of the cover at the end, so you can see if it lies flat and even.
7. Both arms are at home with the grippers open.
8. Both arms reach the folded duvet, the folded cover, both far corners of the mattress, and the middle of the mattress. They reach all of them without stretching to a limit.
9. Both arms slide fabric along the mattress at mattress height. Neither arm lifts a load up above the headboard.
10. The two arms do not hit each other. One arm holding a far corner stays out of the other arm's path along the bed.
11. Nothing hangs off the arms that can catch on the fabric.
12. If the arms cannot reach a place, park the base again with the Base positioning steps until lines 8 to 11 are all true.

Materials checklist

1. The bed is one mattress on one frame. Nobody moves it, leans on it, or sits on it.
2. The mattress has a fitted sheet on it. The sheet is smooth and tucked. Nothing else lies on the mattress.
3. The mattress is clear at the start. No pillows, no blanket, no clothes.
4. The duvet is one duvet that fits the cover. It is dry, clean, and has no tears.
5. The duvet lies folded in a flat square at the foot of the bed, on the mattress, not on the floor.
6. The cover is one duvet cover that fits the duvet. It is dry, clean, right side out, and turned the right way round.
7. The cover lies folded flat next to the duvet at the foot of the bed, with its opening toward the foot end.
8. The cover closes with buttons along its opening. All the buttons are open at the start.
9. The cover has no tears, and no button is missing or loose.
10. The two far corners of the cover are the head corners. They are the corners at the closed end, farthest from the opening.
11. The duvet has no cover on it at the start.
12. There is a clear strip of floor along the side of the bed where the base parks. It is empty and dry.
13. Nothing hangs over the side of the bed at the start.
14. The room floor is clean and dry, so fabric that touches it can be told from fabric that has not.

Workspace layout

Nothing is marked or taped out. You judge every place below by eye, from the bed itself.

Mattress: the whole work surface. All the work happens on it. Both grippers.

Head end: the end of the mattress at the headboard. The two far corners are here. The duvet is pushed toward this end.

Foot end: the end of the mattress away from the headboard. The duvet and the cover start here. The opening of the cover ends up here.

Far corners: the two corners of the mattress at the head end. The matched corners are held over them.

Duvet spot: the flat square of folded duvet at the foot of the bed. It empties as the episode runs.

Cover spot: the flat square of folded cover next to the duvet. It empties as the episode runs.

Bed frame and headboard: never moved, never leaned on, never used to hold fabric.

Floor: out of bounds. No fabric ever touches it.

Vocabulary

Cover opening: the open end of the cover, where the buttons are. The duvet goes in here.

Far corners: the two corners at the closed end of the cover, and the two matching corners of the duvet. These are matched first.

Corner matched: one duvet corner sits right inside one cover corner, tip into tip, with no slack between them, and both are held in one closed grip.

Worked through: the duvet is pushed along inside the cover until the far corners are at the head end and the duvet reaches the opening at the foot end.

Shake: you take the cover at the opening with both grippers, lift it just off the mattress, and move it up and down twice so the duvet drops into the far corners.

Seated: the duvet fills all four corners of the cover, and the cover has no empty flat corner left.

Buttons closed: every button on the opening is pushed through its hole. No button is left open. No hole is left empty.

Flat and even: the cover lies on the mattress with no lump, no fold, and no twist, and it hangs the same amount over both long sides.

Square on the bed: the head end of the cover is level with the head end of the mattress, and the long edges of the cover run parallel to the long edges of the mattress.

Lost grip: the fabric slips out of a gripper before the step is done. You take it again at the same place.

Twist: the cover is turned inside itself, so the duvet lies across the cover instead of along it. A twist is fixed by pulling the duvet back out and starting again at the far corners.

On the floor: any part of the duvet or the cover touches the floor. This ends the episode.

Back and clear: the arm is pulled back, so no part of it is over the mattress or the fabric.

Steps

Step 1: Lay the duvet and the cover out flat

Goal: the duvet lies flat along the bed, and the cover lies flat on top of it with its opening at the foot end.

1.1 Open the duvet out

Close a gripper on the near edge of the folded duvet.

Pull it along the mattress toward the head end until the duvet is open and flat.

Straighten it so its long edges run along the long edges of the mattress.

Check: the duvet lies flat on the mattress, along the bed, with no fold left in it. No part of it hangs on the floor. If it lies across the bed, pull it round until it lies along the bed.

1.2 Open the cover out on top

Close a gripper on the near edge of the folded cover.

Pull it along the mattress on top of the duvet until the cover is open and flat.

Turn it so its opening faces the foot end and its closed end faces the head end.

Check: the cover lies flat on top of the duvet, right side out, opening at the foot end, buttons open. If the opening faces the head end, pull the cover round and lay it down again.

Expected state: the duvet lies flat on the mattress. The cover lies flat on top of it, opening at the foot end. Both grippers are back and clear.

Step 2: Match the two far corners

Goal: each far corner of the duvet sits right inside the matching far corner of the cover, and both are held.

2.1 Reach inside to the first far corner

Open the cover opening with one gripper and hold it open.

Slide the other gripper inside the cover, along the inside, all the way to one far corner.

Close that gripper on the far corner of the cover from the inside.

2.2 Take the duvet corner with it

Bring the matching far corner of the duvet up to the gripper that holds the cover corner.

Push the duvet corner into the cover corner, tip into tip, until there is no slack between them.

Close the gripper on both together and hold them.

Check: one duvet corner sits inside one cover corner, tip into tip, and both are held in one closed grip. If there is slack, open the grip, push the duvet corner in further, and close again.

2.3 Match the second far corner the same way

Do 2.1 and 2.2 again at the other far corner, with the other gripper.

Hold both matched corners.

Check: both far corners are matched, tip into tip, and each is held. The cover is not turned inside itself. If the cover is twisted, let both corners go, pull the cover flat again, and start again at 2.1.

Expected state: both far corners of the duvet sit inside both far corners of the cover. Both grippers hold a matched corner. The rest of the duvet still lies on the mattress toward the foot end.

Step 3: Work the fabric through

Goal: the duvet is all the way inside the cover, from the far corners at the head end to the opening at the foot end.

3.1 Push the matched corners to the head end

Keep both matched corners held.

Slide both grippers along the mattress toward the head end, taking the corners with them.

Stop when both matched corners lie over the two far corners of the mattress.

Check: both matched corners are at the head end of the mattress. Neither corner has slipped out. If a corner slips out, stop, take it again at the same place, and push it back in tip into tip.

3.2 Feed the rest of the duvet along

Open one gripper. Close it on the cover a hand's width back from the head end.

Pull the cover down along the duvet toward the foot end, a hand's width at a time.

Swap hands each pull, so one gripper always holds the fabric still while the other pulls.

Keep the fabric on the mattress. Do not lift the load up in the air.

Check: the cover moves down over the duvet a little at a time and the duvet stays inside it. Nothing hangs on the floor. If a pull drags the duvet along instead of the cover, stop, hold the duvet with the other gripper, and pull again.

3.3 Bring the opening down to the foot end

Keep pulling until the opening of the cover reaches the foot end of the duvet.

Check that the two near corners of the duvet sit inside the two near corners of the cover.

Push each near corner in by hand if it sits short.

Check: the duvet reaches all four corners of the cover, and no part of the duvet hangs out of the opening. If a near corner sits short, push it in until it fills the corner.

Expected state: the duvet is inside the cover, head end to foot end. The opening is at the foot end and still open. Both grippers are back and clear.

Step 4: Shake it to seat the duvet

Goal: the duvet is seated in all four corners of the cover, with no empty corner and no lump.

4.1 Take the cover at the opening

Close both grippers on the cover at the foot end, one near each corner of the opening.
Hold the opening closed in the grips, so nothing falls out.

4.2 Shake it twice

Lift the foot end just off the mattress. Lift it no higher than the mattress top.
Move it up and down twice, so the duvet drops along inside the cover toward the head end.
Lower it back down onto the mattress.
Keep the head end on the mattress the whole time. Do not lift the whole load up in the air.

Check: the duvet has dropped into the far corners. The head end of the cover is full, not flat and empty. If a far corner is still empty, take the cover at the opening again and shake twice more. Do no more than two extra shakes.

4.3 Smooth it down

Run a gripper along the top of the cover from the head end to the foot end, flat and gentle.
Push out any lump you feel toward the nearest edge.

Check: the duvet is seated in all four corners. The cover has no big lump and no fold caught inside it. If the duvet lies across the cover instead of along it, pull the duvet back out through the opening and start again at Step 2.

Expected state: the duvet is seated in the cover. The opening is at the foot end, still open. Both grippers are back and clear.

Step 5: Close the buttons

Goal: every button on the opening is closed, and the opening lies flat.

5.1 Line the opening up

Take the two flaps of the opening, one in each gripper.
Lay them over each other so the buttons line up with their holes.
Lay the opening flat on the mattress.

Check: each button sits next to its own hole. No flap is folded under. If the flaps are out of line, lift them apart and lay them over again.

5.2 Close each button in order

Start at one end of the opening and work to the other end.

Hold the hole open with one gripper. Push the button through the hole with the other gripper.

Do the next button the same way, and keep going to the last one.

Do not skip a button and do not go back.

Check: every button is through its own hole. No button is left open and no hole is left empty. If a button will not go through, hold the hole wider and push it again. Try no more than three times on one button.

5.3 Look along the closed opening

Run a gripper along the closed opening, flat and gentle.

Push the opening flat against the mattress.

Check: the opening lies flat and closed, with no gap between the buttons and no corner of the duvet poking out. If a corner pokes out, open the nearest button, push the corner back in, and close the button again.

Expected state: the cover is closed with all its buttons. The duvet is inside it. Both grippers are back and clear.

Step 6: Spread it on the bed

Goal: the cover lies square and flat over the whole mattress, hanging the same amount over both long sides.

6.1 Square it to the mattress

Close a gripper on the cover at the head end.

Pull it up the bed until the head end of the cover is level with the head end of the mattress.

Straighten it so the long edges of the cover run along the long edges of the mattress.

Check: the head end of the cover is level with the head end of the mattress, and the cover is not turned across the bed. If it sits low or turned, pull it up and round again.

6.2 Even the two sides

Take the cover at the far long edge and pull it out until it hangs over that side.

Take it at the near long edge and pull it out until it hangs over that side the same amount.

Look along the bed from the side to see that both hangs look the same.

Check: the cover hangs the same amount over both long sides. If one side hangs much longer, pull the cover across toward the short side until the two look the same.

6.3 Smooth it flat

Run a gripper over the top of the cover from the head end to the foot end.

Push out every fold and lump toward the nearest edge.

Do not lift the cover off the mattress to do it.

Check: the cover lies flat and even, with no fold, no lump, and no twist. The whole mattress is covered. If a fold is left, smooth that spot again.

Expected state: the duvet is inside the closed cover, and the cover lies flat, square, and even over the whole mattress.

Step 7: End the episode

Check the duvet: it is inside the cover, seated in all four corners, with no part of it out of the opening.

Check the cover: it is closed at every button, right side out, and not twisted.

Check the spread: the head end is level with the head end of the mattress, the cover lies flat, and both sides hang the same.

Check the floor: no part of the duvet or the cover touches it.

Check the bed: the mattress and the frame stand where they started, and the sheet under the cover is not pulled loose.

Check the base has not moved.

Fix any failed check before you end.

Send both arms home with the grippers open. Then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Open every button on the cover opening.
2. Pull the duvet out of the cover through the opening.
3. Shake the duvet out and fold it into a flat square. Lay it at the foot of the bed on the mattress.
4. Turn the cover right side out if it came out inside out. Fold it into a flat square. Lay it next to the duvet, opening toward the foot end, buttons open.
5. Smooth the fitted sheet on the mattress and tuck it back in at every corner.
6. Clear the mattress. Nothing lies on it but the folded duvet and the folded cover.
7. Change the cover if a button is missing, a button hole is torn, or the fabric is torn.
8. Change the duvet if it is torn, damp, or too big for the cover.
9. Change either one if it touched the floor during the episode. Send it to the wash.
10. Check the floor strip along the side of the bed is clear and dry.
11. Check the base is still locked and parked square. Then do the Base positioning steps and both Setup checklists again.

SOP violations

These are actions that break the SOP. They are reviewed side by side in the review tool.

How to record a violation in review

For each violation, write down the start timestamp, the violation name, and the SOP rule broken. The visible cue is what the reviewer sees. The coaching note is for retraining. It is not an annotation label.

Episode handling

Tag every violation with its timestamp and its name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Base moved during the episode

Visible cue: the bed moves in the frame, the mattress edge changes angle in the frame, or the base rolls, slides, or turns at any time after recording starts.

SOP rule broken: Steps 1 to 7. The base is parked and locked before recording and stays still for the whole episode.

Coaching note: park it, lock it, push it to test it, then start recording. A base that moves is a lost episode.

Violation: Fabric touched the floor

Visible cue: any part of the duvet or the cover falls off the bed and touches the floor at any time.

SOP rule broken: Steps 1 to 6. Nothing ever touches the floor.

Coaching note: keep the load on the mattress. If it is sliding off, hold it lower down, not further out.

Violation: Lifted the load up in the air

Visible cue: the grippers lift the whole duvet or the whole cover clear of the mattress and hold it up, instead of sliding it along the mattress.

SOP rule broken: Steps 1.1, 1.2, 3.1, 3.2, and 4.2. All the work stays on the mattress, and only the foot end lifts, and only for the shake.

Coaching note: slide it, do not carry it. The only lift in this task is the shake at the foot end.

Violation: Corners not matched first

Visible cue: the duvet is pushed into the cover before both far corners are matched, or only one far corner is matched before the fabric is worked through.

SOP rule broken: Step 2. Both far corners are matched, tip into tip, before anything is worked through.

Coaching note: two corners first, every time. A duvet that goes in without its corners comes out crooked.

Violation: Corner slipped and was not retaken

Visible cue: a matched corner comes out of a gripper during Step 3 and the work carries on without taking it again.

SOP rule broken: Steps 2.2, 2.3, and 3.1. A matched corner is held until the fabric is through, and a lost grip is taken again at the same place.

Coaching note: stop the moment it slips. Two seconds now saves a full restart later.

Violation: Cover put on twisted

Visible cue: the duvet ends up lying across the cover instead of along it, or the cover is turned inside itself, so one corner of the duvet is nowhere near a corner of the cover.

SOP rule broken: Steps 2.3 and 4.3. The cover lies straight over the duvet, and a twist is fixed by pulling the duvet out and starting again at the far corners.

Coaching note: check the cover is straight before you pull. You cannot smooth a twist out from the top.

Violation: Cover inside out or the wrong way round

Visible cue: the cover is laid out with its seams showing, or with its opening at the head end instead of the foot end.

SOP rule broken: Step 1.2. The cover is right side out, opening at the foot end.

Coaching note: find the opening first. It always goes at the foot.

Violation: Duvet not seated in the corners

Visible cue: at the end of Step 4, a corner of the cover is flat and empty, or the duvet ends short of one end.

SOP rule broken: Steps 3.3 and 4.2. The duvet fills all four corners of the cover.

Coaching note: shake it twice, then look at all four corners. An empty corner shows straight away.

Violation: Shaken too hard or too high

Visible cue: the foot end is lifted well above the mattress top, the load is swung, or the shake throws the fabric off the bed.

SOP rule broken: Step 4.2. Lift the foot end just off the mattress, no higher, and move it up and down twice.

Coaching note: small and twice. A big swing puts the duvet on the floor.

Violation: Buttons left open or skipped

Visible cue: at the end of Step 5 a button is still open, a hole is empty, or the buttons are done out of order with one missed in the middle.

SOP rule broken: Step 5.2. Every button is closed, in order, from one end to the other.

Coaching note: one end to the other, no skipping. Run back along the line before you call it closed.

Violation: Duvet poking out of the opening

Visible cue: a corner or an edge of the duvet sticks out through the closed opening at the end of Step 5.

SOP rule broken: Step 5.3. No corner of the duvet pokes out of the closed opening.

Coaching note: push it in, then close the button. Do not close a button over a corner.

Violation: Leaned on or moved the bed

Visible cue: a gripper, wrist, or forearm rests on the bed frame or the headboard. Or the mattress slides on the frame. Or the fitted sheet is pulled loose.

SOP rule broken: Steps 1 to 6. Nobody leans on, pushes, drags, or lifts the bed frame or the mattress.

Coaching note: the arm holds itself up. If the bed moves when you pull, you are pulling from the wrong place.

Violation: Both hands pulled at once

Visible cue: both grippers pull the fabric at the same time in Step 3, so nothing is held still.

SOP rule broken: Step 3.2. One gripper holds the fabric still while the other pulls, and the hands swap each pull.

Coaching note: one holds, one pulls, then swap. Two pulls at once just drags the whole bed.

Violation: Done in the wrong order

Visible cue: the buttons are closed before the duvet is seated, the cover is spread before the buttons are closed, or the shake happens before the fabric is worked through.

SOP rule broken: Steps 1 to 6. Lay out, match the corners, work it through, shake, close the buttons, spread.

Coaching note: the order is the task. Every step out of order has to be undone.

Violation: Spread crooked or uneven

Visible cue: at the end of Step 6 the head end of the cover sits well short of the head end of the mattress, the cover runs across the bed, or one side hangs far longer than the other.

SOP rule broken: Steps 6.1 and 6.2. The cover is square to the mattress and hangs the same amount over both sides.

Coaching note: head end first, then even the sides. Look along the bed from the side before you stop.

Violation: Folds and lumps left in

Visible cue: at the end of Step 6 the cover shows a big fold, a lump, or a ridge across the top.

SOP rule broken: Steps 4.3 and 6.3. Smooth the cover flat from the head end to the foot end.

Coaching note: push every lump out to the nearest edge. Lumps do not settle on their own.

Violation: Something dropped or knocked off the bed

Visible cue: the duvet or the cover slips out of a gripper and falls off the side of the bed, or is knocked off by an arm going past.

SOP rule broken: Steps 1 to 6. One load moves at a time, and each one is put down under control.

Coaching note: close on the fabric all the way before you pull, and put it down before you let go.

Violation: Failed check not retried

Visible cue: a check in a step clearly fails, and the episode carries on with no retry. For example, a corner sits short, a far corner is empty after the shake, or a button is left open.

SOP rule broken: Steps 1 to 6. Do the retry written under each failed check before you move on.

Coaching note: a check is only worth doing if you do the retry. Fix it where it happened.

Violation: Wrong ending

Visible cue: the episode ends with the duvet part out of the cover, a button open, the cover crooked on the bed, or fabric on the floor. Or an arm is not home, or a gripper is not fully open.

SOP rule broken: Step 7. Check the duvet, the cover, the spread, the floor, the bed, and the base. Then send both arms home with the grippers open and stop recording.

Coaching note: check first. Going home is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, throw the episode away, and never use them for coaching.

Recording stopped or paused during the episode (recording system).

Camera dropped frames or lost its feed (capture system).

Camera cannot see the far corners, the opening, or the buttons, so nobody can tell if the corners were matched or the buttons were closed.

Hardware fault on an arm: gripper failure, drift, a crash caused by the controller, or a motor error.

Base fault: a lock that will not hold, or a base that rolls, slides, or turns with the lock on.

Faulty cover: one with a missing button, a torn button hole, a torn seam, or an opening that will not lie flat when closed.

Faulty duvet: one that is torn, damp, lumpy inside, or too big to go into the cover at all.

Mismatched set: a duvet and a cover that are not the same size, so the corners cannot be matched.

Faulty bed: a mattress that slides on its frame under a correct pull, or a frame that rocks.

Faulty sheet: a fitted sheet that comes off its corners under a correct pull, so the mattress cannot be worked on.

A place turns out to be too far for the arms with the base parked correctly, so a far corner of the mattress, the folded duvet, or the folded cover cannot be reached without stretching or folding the arm.

Annotation subtasks (from SOP)

1. Open the duvet out flat on the mattress
2. Open the cover out flat on top of the duvet
3. Match the first far corner
4. Match the second far corner
5. Push both matched corners to the head end
6. Feed the cover along the duvet
7. Bring the opening down to the foot end
8. Take the cover at the opening
9. Shake the duvet into the far corners
10. Smooth the duvet down inside the cover
11. Line the opening up
12. Close each button in order
13. Square the cover to the mattress
14. Even the two sides
15. Smooth the cover flat
16. Send both arms home and end the episode